Lb-MEC

Reg. No. | | | | | | | | | | | **[18BS3MA01]**

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

MATHEMATICS-III

Time: 3 Hours **Maximum Marks: 70**

Instructions:

i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Find the coefficient of correlation and regression lines to the following data: **07**

x	5	7	8	10	11	13	16
y	33	30	28	20	18	16	9

b. Using least-squares method, fit a curve of the form $y = ae^{bx}$ to the data **07**

x	1	2	3	4	5	6
y	7.209	5.265	3.846	2.809	2.052	1.499

2. a. Given that $f(x) = x + x^2$ for $-\pi < x < \pi$, find the Fourier expression of $f(x)$. **07**

b. Obtain the constant term and the coefficient of the first sine and cosine terms in the Fourier series of $f(x)$ as given in the following table. **07**

x	0	1	2	3	4	5
$f(x)$	9	18	24	28	26	20

3. a. Find the Fourier transform of the function $f(x) = \begin{cases} x^2, & |x| \le a \\ 0, & |x| > a \end{cases}$ where 'a' is a positive constant. **07**

b. Find the Fourier sine transform of $f(x) = e^{-|x|}$ and hence evaluate $\int_0^\infty \frac{x \sin ax}{1+x^2} dx$ where 'a' is a positive constant. **07**

OR

4. a. Find the Z-Transforms $\cos n\theta$ and hence find $Z(n\cos n\theta)$. **07**

b. Using Z-transform, solve $u_{n+2} + 4u_{n+1} + 3u_n = 3^n$ with $u_0 = 0,\ u_1 = 1$. **07**

5. a. Solve $\frac{\partial^2 z}{\partial x \partial y} = x^2 y$ subject to the conditions $z(x,0) = x^2$ and $z(1,y) = \cos y$. **07**

b. Solve: $(x^2 - yz)p + (y^2 - zx)q = z^2 - xy$. **07**

OR

6. a. Derive one dimensional Wave equation. **07**

b. Solve by the method of separation of variables the partial differential equation

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}.$$ **07**

7. a. Use Newton-Raphson method, find the real root of $x \log_{10} x = 1.2$ correct to correct to four decimal places near to 2. **07**

b. Use suitable interpolation to find the equation of the cubic curve which passes through the points (4, -43), (7, 83), (9, 327) and (12, 1053). **07**

OR

8. a. Use Weddle's rule to evaluate $\int_0^1 \frac{dx}{1+x}$ and hence find approximate value of $\log_e 8$. **07**

b. Using the Runge-Kutta method, find the solution of the problem $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ at the point $x = 0.1$ with h=0.1. **07**